

I work on solutions for secure Ministry of Defence (MoD) environments, helping automate Active Directory onboarding, Kerberos and SPNEGO authentication, SMB storage, Group Policy, Folder Redirection and Joiner/Mover/Leaver (JML) workflows using Python and PowerShell.

Over the years I've also worked on Android system development, Mobile Device Management (MDM), kiosk lockdown solutions, Device Policy Manager integrations, custom APK update servers, digital signage platforms, screencasting technologies, handwriting-recognition systems and reverse-engineering activities for analysis, troubleshooting and security investigations.

Work Experience

2025–2026 Current	Software Developer — Akhter Computers Identity, Active Directory and Infrastructure Automation - Created an identity-driven prototype connecting Windows authentication, Kerberos/SPNEGO validation, backend token services and protected file access workflows. - I also made a Windows Single Sign-On flow where a domain user logs into Windows, obtains Kerberos tickets, negotiates with a Python FastAPI identity service, receives an RS256 JWT, and accesses backend services through JWKS-validated trust. - Worked with Samba Active Directory and Windows Server 2022 lab environments to validate AD, DNS, Kerberos, domain logon and Windows 11 Pro domain join behaviour. - Moved the project direction toward an enterprise-native AD/SMB/GPO automation toolkit: central user storage through SMB, Folder Redirection through Group Policy, and Python/PowerShell orchestration instead of custom file synchronisation. - To help IT Operations I made a Pairing-Based Domain Onboarding Tool for guided server/client onboarding, including pairing flow, client DNS configuration, domain join workflow and validation-oriented state-machine planning. - Created packaging and deployment workflows for Python services using Debian packages, virtual environments, systemd services, environment files and hardened runtime configuration. - Investigated Kerberos operational issues including SPNs, keytab generation, GSSAPI validation, service principal mapping, replay-cache permissions and Windows client ticket behaviour. - Maintained technical documentation and runbooks covering KVM lab setup, Windows Server/Windows 11 testing, Samba AD recovery, DNS/SRV troubleshooting, Kerberos pitfalls and deployment procedures.
2024–2025	Android System Developer — Distributed Device Systems - Built distributed digital-signage and device-synchronisation features with a local-first architecture for environments where cloud connectivity may be limited. - Worked on endpoint orchestration, device communication and resilience patterns later applied to identity-aware infrastructure and onboarding systems.
2023–2024	Android System Developer — Remote MDM & Update Infrastructure - Developed Android endpoint-control and update infrastructure, including secure APK delivery, remote management workflows, signature/hash verification and controlled command execution. - Implemented remote device management concepts across Android clients, Linux-backed services and operational tooling.
2021–2023	Software Development Apprenticeship - Contributed to IoT and Android projects involving telemetry, gateway communication, PostgreSQL-backed monitoring and field diagnostics. - Built a foundation in systems development, debugging and customer-facing operational support.

SKILLS

Identity

Active Directory
Kerberos / SPNEGO
SSO / JWT / JWKS
RBAC / JML

Windows

Server 2022
SMB / NTFS
GPO
Folder Redirection

Automation

Python
PowerShell
FastAPI
Orchestration

Endpoint

Android MDM
Kiosk
Debian Packaging
Secure Endpoint Mgmt

Lab

KVM
qcow2
Reproducible Testing
LaTeX

Main Projects



Pairing-Based Domain Onboarding Platform

Built a guided onboarding platform for Windows Server and Active Directory environments, covering server/client pairing, DNS configuration, domain-join validation, SMB home-directory setup, Group Policy and Folder Redirection workflows.

AD / GPO



Windows SSO / Kerberos to JWT Prototype

Designed a working authentication flow where Windows domain login is reused to access protected backend services through Kerberos/SPNEGO validation and RS256 JWT trust.

SSO



Android MDM, Kiosk & Secure Endpoint Control

Developed controlled Android endpoint solutions including kiosk lockdown, Device Policy Manager integration, secure update infrastructure and administrator-authorised privileged actions.

MDM



Distributed Digital Signage Platform

Worked on Android-based digital-signage systems focused on content delivery, remote device management, update distribution, local-first behaviour and resilience with limited connectivity.

Signage